Honours and MInf Projects

Dissertation / How to write

Writing the dissertation

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> For additional guidance on writing and other resources, please see the Resources page.

Style Guide

The `infthesis` documentclass is provided for writing project reports in LaTeX.

The report template zip file (shown below) contains the `infthesis.cls`, a skeleton file for the UG project report (`skeleton.tex`) and for the bibliography (`mybibfile.bib`) and associated images files to include the UoE shield. It also contains a `ugcheck.sty` file that performs automatic checking when compiling LaTeX that the document meets the required formatting conditions.

To use the template

- Download the report template zip file.
- Unzip it in a local directory on DICE, or on your own machine (assuming you have LaTeX installed) or upload it as a new project on Overleaf (external link).
- Look at the README file for advice on how to compile the document.
- Use `skeleton.tex` and `mybibfile.bib` as the template for your project.

Note that the same template should also be used for MInf1 and MInf2 projects.

Overall format

The report must contain **no more than 40 pages** (and no less than 20 pages) of content (that is, 40 pages from the start of the Introduction to the end of the Conclusion).

The report should include:

- Title page with abstract (i.e., a one or two paragraph summary of the contents).
- Introduction and background synopsis, in which the project topic is described and set in the context of published literature, and the main results are briefly summarized (about five pages).
  - **A clear and concise summary of your contributions** (examples: adapting a suite of existing code; interpreting a theoretical algorithm; coding; testing; conducting an experiment) preferably as a bulleted list.
  - **MInf 2 only**: This chapter should include a section entitled Previous work carried out in which you give a brief account of what you did last year. This should consist of what the reader needs to know so that he/she can place the new work in context. It will also make the new achievements clear. Note, in particular, that you cannot assume that the reader of this report has read the report for Part 1 of your project. For that reason you may well want to repeat material from that report covering background, related work, etc., updating and adjusting as appropriate.
- Discussion of the work undertaken, in which the various sub-problems, solutions and difficulties are examined. Discussion of any implementation should concentrate on overall principles and aspects of particular interest, rather than minute details. You should always make it clear what was completed, and what was left as future work. If in doubt spell it out: tell us which bits of code you could find already written, which bits you had to do from scratch, which bits were routine, which bits were challenging (and why).
- If appropriate, a description of experiments undertaken, a presentation of the data gleaned from them, and an interpretation of that data.
- **MInf 1 only**: A plan for Part 2 of the project, in which a detailed explanation is given of specific goals for building during the second year of the project on the work done during the first year. NB: if you are unsure about continuing the project, this can be opted out.
- Conclusion, in which the main achievements are reviewed, and unsolved problems and directions for further work are presented.
- Bibliography.

In addition, the report sometimes contains appendices in which relevant program listings, circuit diagrams, check plots, formal proofs etc. are included. Such appendices do not contribute to the page count but they should not be excessively long.

Excessive length, repetition and rambling explanations are discouraged. Try to write in a brief and concise but clear style. On balance, information about what you have done and choices you have made etc. is much more important than general background information.

Please use a spell-checker.

It is perfectly acceptable, and indeed recommended, that you arrange for your supervisor to read and comment on a draft of the project report well in advance of the deadline. Ideally you

would have a full draft ready at least two weeks in advance, so you have plenty of time to make any modifications that your supervisor suggests.

Background chapter

The aim of writing a draft background chapter by Week 5 in Semester 1 is to give you early feedback on the following assessment criteria:

- Understanding of the problem
- Knowledge of the literature
- Critical evaluation of previous work
- Quality of the report

Knowledge of the literature (previously published work relevant to your project) and critical evaluation of previous work (which requires understanding of the problem) are usually evidenced in your dissertation through a background chapter. Sometimes the information is distributed differently, for example, in an extended introduction and synopsis; or with details of a paper that you are following very closely explained in the methods. Either way, writing some of this content now, however it finally ends up in your dissertation, will also be a good test for the quality of your writing.

What is required?

You should aim to produce around 5 pages of synopsis of the existing work that forms the context for your project. You should give this to your supervisor by the end of week 5, and discuss it with them at your next meeting, jointly filling in and submitting this form (MS Forms).

It is suggested that you use this as an opportunity to familiarise yourself with LaTeX (external link), using the report template shown in Style Guide provided. The University has reasonable guides for beginners on how to use LaTeX with or without overleaf.

Things to consider:

- Although typically published papers will be a substantial part of the background (see more below) this may differ for different projects. For example, your task may instead require you to get a strong understanding of an existing code-base, or critical software tools, or of an application domain. You will still need to demonstrate your knowledge and critical evaluation through writing a good synopsis of the relevant existing material, to set your work in context, and justify your approach.

- A good background chapter is not just a summary of all the things you have read. Rather, it should be selective and well-organised, to tell the reader just what they need to know to understand the work that follows.

- It should be clear to the reader (and to you!) why you are including any information that you present. A lengthy, rambling and 'padded' background chapter is a common reason for lower marks on these criteria; or a chapter that omits critical material.
- Probably the best guide to the expected content is to look at published papers in the same domain as your project, and how they put their new work in the context of previous work.
- **Note for MInf 2:** your MInf 1 dissertation forms part of the background literature, and you can refer to it as such. This does not mean you should assume the marker has read it. Rather, you should summarise it (and other relevant material) sufficiently clearly that the reader has the context for understanding your current work. Looking at a published paper that similarly builds directly on previous work from the same authors should be a good guide.

Researching the background

Your project supervisor should have pointed you towards some key sources for your project, but is expected (certainly to achieve an excellent grade) that you will put some significant effort into going beyond those sources.

There are many literature search tools you can use, but using Google Scholar as an example, here are some tips, starting from a key reference:

- 'Cited by' will show you all the subsequent papers that refer to your source paper. So if anyone has followed up the work already, you should find it here.
- 'Related articles' show you all the papers that have overlapping references to your source paper. The top hits are very likely to be addressing similar problems.
- The source paper itself will cite other papers, and you should be able to tell from the paper which of these might be crucial (e.g. another paper that was the first presentation of an important method that the current paper uses) and choose which you should read first-hand.
- If you have some key terms, try searching for those terms plus 'review' or 'survey', and limiting the search to recent dates. This can be a good way to find if there is a recent published overview of the topic. Such a paper is also likely to contain references to other useful papers.

Bibliographies

The bibliography of your report is a list of all of the books, articles etc. used in the project and referred to in the report. There are official standards for these things, but here are some examples that cover most references required in reports.

- A book:
  W. Wang. Beginning Programming For Dummies, 4th edition. John Wiley & Sons, 2006.
- A journal article:
  S. Brin and L. Page. The anatomy of a large-scale hypertextual web search engine. Computer Networks 30(1-7):107-117, 1998.
- A conference paper:
  O. Kammar and G. Plotkin. Algebraic foundations for effect-dependent optimisations. In Proc. 39th ACM SIGPLAN-SIGACT Symp. on Principles of Programming Languages, POPL 2012, pages 349-360, 2012.
- A source on the web that you can't fit into any other category:
  Wikipedia. Hash table. http://en.wikipedia.org/wiki/Hash_table, accessed 10 Oct 2013.

The author of an item on the internet - as in the last example above - is often unclear, so please just do the best you can. In such cases, using the publication venue (Wikipedia, Stack Overflow, etc.) is a reasonable solution since then at least these items will appear together in your list of references.

The exact presentation of the information is not nearly as important as providing enough information and using a consistent format. You must provide enough information to enable the reader to find exactly the item you are referring to without guesswork or search. If you can add information to make it easier to do that, then please do so. You should also indicate what the item is without forcing the reader to look it up. So don't just give a URL without indicating title, author (if known), etc. It is easy to get this right, and failure to do so is a sign of sloppiness that the reader may suspect extends to other aspects of your work.

You can cite items in the bibliography in the style "In Brin and Page (1998), ..." but it is more typical in Informatics to give every item in the bibliography a unique identifier which is used to refer to it in the text. If you use LaTeX bibliographies (BiBTeX with the `\cite` command) then this happens automatically and you can choose between numbered citations like [17] (`bibliographystyle{plain}`) and author-initials-plus-year citations like [ST93] (`\bibliographystyle{alpha}`), etc. with automatic sorting of entries.

DBLP (external link) is a very useful resource for finding the information you need to include in bibliography entries. Once you have found an item, you can export the BiBTeX entry using the DBLP download icon.